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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 3.

LIMITS II

Uniqueness theorem for limits
Main limit theorem (Limit properties)
Squeeze theorem
3. Homework

Uniqueness theorem for limits

Theorem 3.1.

If the limit of function f(x) exists at point c, then it is unique.

Proof (by contradiction)

Lets say $\lim_{x\to c} f(x) = \mathrm{A} \neq B = \lim_{x\to c} f(x)$

Lets say $|A - B| = 2\varepsilon > 0$

From existance of $\lim_{x\to c} f(x) = \mathrm{A}$ follows that

$\varepsilon > 0 \exists \delta_1 > 0 x \in Dom(f)[0 < |x - c| < \delta_1 \Rightarrow |f(x) - A| < \varepsilon]$.

From existance of $\lim_{x\to c} f(x) = \mathrm{B}$ follows that

$\varepsilon > 0 \exists \delta_2 > 0 x \in Dom(f)[0 < |x - c| < \delta_1 \Rightarrow |f(x) - B| < \varepsilon]$.

Lets select $\delta = \min\{\delta_1, \delta_2\}$.

Then $\forall x \in Dom(f)[0 < |x - c| < \delta \Rightarrow$

$$2\varepsilon = |A - B| = \left|\left(A - f(x)\right) + (f(x) - B)\right| \leq$$

$$\leq |f(x) - A| + |f(x) - B| < \varepsilon + \varepsilon = 2\varepsilon].$$

The obtained contradiction $2\varepsilon < 2\varepsilon$ proves that the assumption is wrong.

The use of the uniqueness theorem for limits is usually associated with the ability to prove that the function f(x) does not exist limit when x approaches the point c. Note that if we obtain different results after calculating limit at the same point, then least one error is made.

Uniqueness theorem for limits

Theorem 3.2.

If $X \subseteq Y$ and $\lim\limits_{\substack{x \to c \\ x \in Y}} f(x) = \mathrm{A}$, then $\lim\limits_{\substack{x \to c \\ x \in X}} f(x) = \mathrm{A}$

Proof

Lets arbitrary choose $\varepsilon > 0$, then according to given: $\exists \delta > 0$ such that $\forall x \in Y \cap Dom(f)[|\mathrm{x} - \mathrm{c}| < \delta \Rightarrow |f(x) - A| < \varepsilon]$.

Then also $\forall x \in X \cap Dom(f)[|\mathrm{x} - \mathrm{c}| < \delta \Rightarrow |f(x) - A| < \varepsilon]$.

Thus,

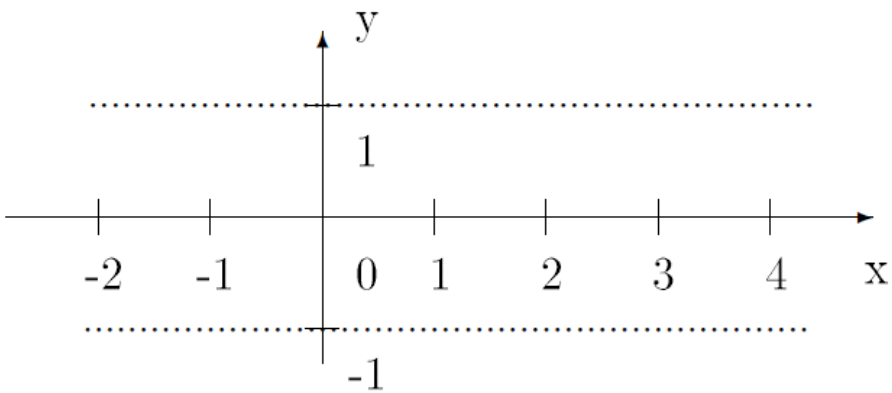
$$\lim_{\substack{x \to c \\ x \in X}} f(x) = \mathrm{A}.$$

Uniqueness theorem for limits

Example

Lets consider Dirichlet Function

$$D(x) = \begin{cases} -1 & , \text{ ja } \quad x \in \mathbb{Q}\,; \\ 1 & , \text{ ja } \quad x \notin \mathbb{Q}\,. \end{cases}$$



As the limit from constant is constant, then

$$\lim_{\substack{x\to 0\\ x\in\mathbb{Q}}} D(x) = -1, but \lim_{\substack{x\to 0\\ x\notin\mathbb{Q}}} D(x) = 1$$

In accordance with theorem 3.2. we can say:
If $\lim_{x\to 0} D(x)$exist, then

$$\lim_{x\to 0} D(x) = \lim_{\substack{x\to 0\\ x\in\mathbb{Q}}} D(x) = -1$$

Similarly we san say:

$$\lim_{x\to 0} D(x) = \lim_{\substack{x\to 0\\ x\notin\mathbb{Q}}} D(x) = 1$$

In turn, the theorem 3.1. makes us conclude that $\lim_{x\to 0} D(x)$ does nit exist.

Main limit theorem (Limit properties)

Proving the existence an values of limits using definition is both time consuming and difficult. We can use several theorems in turn.

Main Limit Theorem

If $\lim\limits_{x\to c} f(x) = A, \lim\limits_{x\to c} g(x) = B, n \in \mathbb{Z}, n > 0, k \in \mathbb{R} - constant$, then

1) $\lim\limits_{x\to c} k = k$
2) $\lim\limits_{x\to c} x = \mathrm{c}$
3) $\lim\limits_{x\to c} k\, f(x) = k \lim\limits_{x\to c} f(x) = kA$
4) $\lim\limits_{x\to c}[f(x) + g(x)] = \lim\limits_{x\to c} f(x) + \lim\limits_{x\to c} g(x) = A + B$
5) $\lim\limits_{x\to c}[f(x) - g(x)] = \lim\limits_{x\to c} f(x) - \lim\limits_{x\to c} g(x) = A - B$
6) $\lim\limits_{x\to c}[f(x)g(x)] = \lim\limits_{x\to c} f(x) \cdot \lim\limits_{x\to c} g(x) = AB$
7) $if \lim\limits_{x\to c} g(x) = B \neq 0, then \lim\limits_{x\to c} \frac{f(x)}{g(x)} = \frac{\lim\limits_{x\to c} f(x)}{\lim\limits_{x\to c} g(x)} = \frac{A}{B}$
8) $\lim\limits_{x\to c}[f(x)]^n = [\lim\limits_{x\to c} f(x)]^n$
9) $\lim\limits_{x\to c} \sqrt[n]{f(x)} = \sqrt[n]{\lim\limits_{x\to c} f(x)}, provided \lim\limits_{x\to c} f(x) > 0\ wnen\ n\ is\ even$

Read as, e.g., «the limit of the sum is the sum of the limits»

Main limit theorem (Limit properties)

Substitution theorem

If f is a polynomial function or a rational function, then $\lim_{x \to c} f(x) = \mathrm{f(c)}$ provided f(c) is defined.In the case of rational function, this means that the value of the denominator at c is not zero.

Example

1. $$\lim_{x \to 2} \frac{3x^2+7x-1}{\sqrt{6x+4}} = \frac{3 \cdot 2^2+7 \cdot 2-1}{\sqrt{6 \cdot 2+4}} = \frac{25}{4}.$$

2. $$\lim_{x \to 1} \frac{5x^2+2x-7}{x^3-1} = \frac{0}{0}$$

Such situations are called indeterminate forms of limits. By using algebraic transformations and arithmetic operations with limits, the indeterminate form can be diverted. In this case wecan shorten because c = 1.

$$\lim_{x \to 1} \frac{5x^2+2x-7}{x^3-1} = \lim_{x \to 1} \frac{5(x-1)(x+\frac{7}{5})}{(x-1)(x^2+x+1)} = \lim_{x \to 1} \frac{5x+7}{x^2+x+1} = \frac{12}{3} = 4.$$

Main limit theorem (Limit properties)

Example

3. $$\lim_{x\to -2} \frac{2-\sqrt{6+x}}{3-\sqrt{7-x}} = \lim_{x\to -2} \frac{(2-\sqrt{6+x})(2+\sqrt{6+x})(3+\sqrt{7-x})}{(3-\sqrt{7-x})(3+\sqrt{7-x})(2+\sqrt{6+x})} =$$

$$= \lim_{x\to -2} \frac{(4-(6+x))(3+\sqrt{7-x})}{(9-(7-x))(2+\sqrt{6+x})} = \lim_{x\to -2} \frac{(-2-x)(3+\sqrt{7-x})}{(2+x)(2+\sqrt{6+x})} =$$

$$= -\frac{3+3}{2+2} = -\frac{3}{2}.$$

(We use multiplication with related expressions)

Main limit theorem (Limit properties)

The main limit theorem remains true for both left- and right-hand limits.

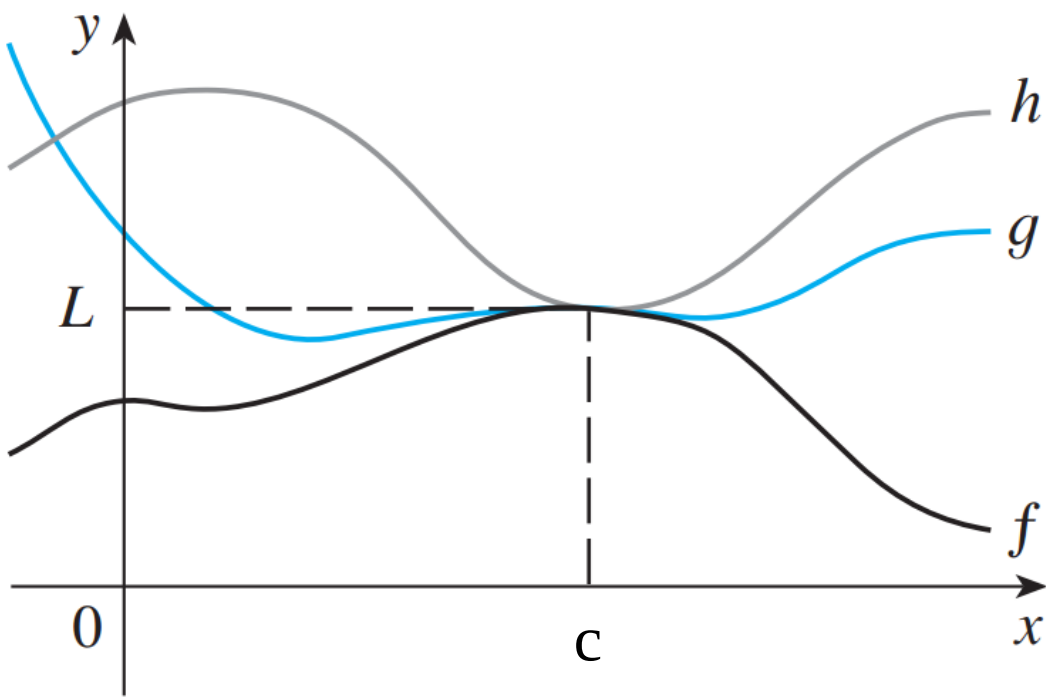
Example

$$\lim_{x\to 2^+} \frac{(x^2+1)\lfloor x\rfloor}{(3x-1)^2} = \frac{5\cdot 2}{25} = \frac{2}{5} \text{ un } \lim_{x\to 2^-} \frac{(x^2+1)\lfloor x\rfloor}{(3x-1)^2} = \frac{5\cdot 1}{25} = \frac{1}{5}.$$

It means that $\lim_{x\to 2} \frac{(x^2+1)\lfloor x\rfloor}{(3x-1)^2}$ does not exist, because one-sided limits differ.

Squeeze theorem

Let f, g, and h be functions statisfiying $f(x) \leq \mathrm{g(x)} \leq \mathrm{h(x)}$ for all x near c, except possibly at x_{0c}. If $\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$, then $\lim_{x \to c} g(x) = L$.



https://tex.stackexchange.com/questions/397786/tikz-illustration-for-the-squeeze-theorem

Squeeze theorem

Example. Lets find $\lim_{x\to 0} sin\frac{1}{x}$!

Separately taking the limit from the first multiplier is 0, but the second – does not exist. But we cannot judge it in such way – the individual limits may not exist, but the limit of prouct can exist. It is also the case here. Note that

$$0 \le \left|xsin\frac{1}{x}\right| \le |x| \cdot 1 = |x|.$$

As $\lim_{x\to 0} 0 = 0\ and \lim_{x\to 0}|x| = 0$, the according to the squeeze theorem

$$\lim_{x\to 0}|x\ sin\frac{1}{x}| = 0.$$

It is easy to proof that

$$(\lim_{x\to c} f(x) = 0) \Leftrightarrow (\lim_{x\to c}|f(x)| = 0).$$

Thus, $\lim_{x\to 0} x\ sin\frac{1}{x} = 0$

HOMEWORK 3

5. Please, find the limits:

$$\text{a)} \lim_{x\to -1} \frac{3x^2+3x}{2x^2-3x-5}$$

$$\text{b)} \lim_{x\to 2} \frac{\sqrt{x^2+5}-3}{\sqrt{2x+5}-\sqrt{11-x}}$$

Answers: a) $\frac{3}{7}$, $b)\frac{4}{3}$

6. Please, find the one-sided limits:

$$\text{a)} \lim_{x\to 0^-} \frac{5x-|x|}{2x}, \qquad \text{b)} \lim_{x\to 0^+} \frac{5x-|x|}{2x},$$

$$\text{c)} \lim_{x\to -1^-} \frac{1}{x-\lfloor x\rfloor}, \qquad \text{d)} \lim_{x\to -1^+} \frac{1}{x-\lfloor x\rfloor}.$$

Answers: a) 3, $b)$ 2, $c)$1, $d)$*does not exist*

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p.68 – 73.
Chapter 1. Section 1.3.